

Given: Due:	Further Maths (Decision) HW 16	Name:
----------------	-----------------------------------	-------

Chapter 4: Route Inspection

You MUST refer to class notes and complete personal study before attempting this homework. Attempt EVERY part of EVERY question. Check your work for errors or omissions before handing in.

91% (A*)	76% (A)	66% (B)	56% (C)	50% (D)	40% (E)
----------	---------	---------	---------	---------	---------

Useful resources: Your notes from recent lessons & your text book

Questions - do these on lined paper, and where appropriate, in the answer booklet extracts.

Q1.

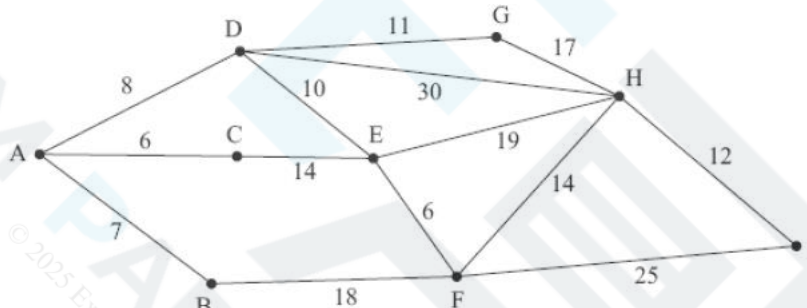


Figure 3

Figure 3 shows a network of roads. The number on each arc represents the length, in km, of that road.

(a) Use Dijkstra's algorithm to find the shortest route from A to I. State your shortest route and its length.

(5)

Sam has been asked to inspect the network and assess the condition of the roads. He must travel along each road at least once, starting and finishing at A.

(b) Use an appropriate algorithm to determine the length of the shortest route Sam can travel. State a shortest route.

(4)

(The total weight of the network is 197 km.)

Q2.

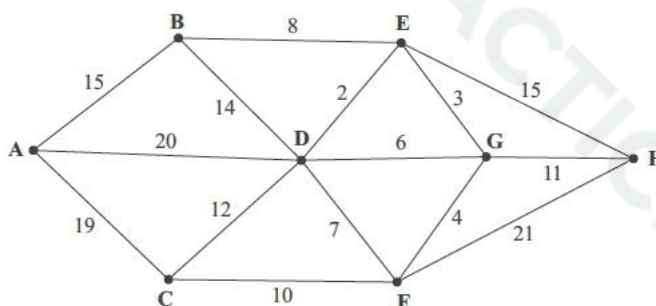


Figure 3

[The total weight of the network is 167]

Figure 3 represents a network of paths. The number on each arc gives the time, in minutes, to travel along that path.

(a) Use Dijkstra's algorithm to find the quickest route from A to H. State your quickest route and the time taken.

(5)

Kevin must walk along each path at least once and return to his starting point.

(b) Use an appropriate algorithm to find the time of Kevin's quickest possible route, starting and finishing at A. You should make your method and working clear.

(5)

Q3.

[The total weight of the network is 451]

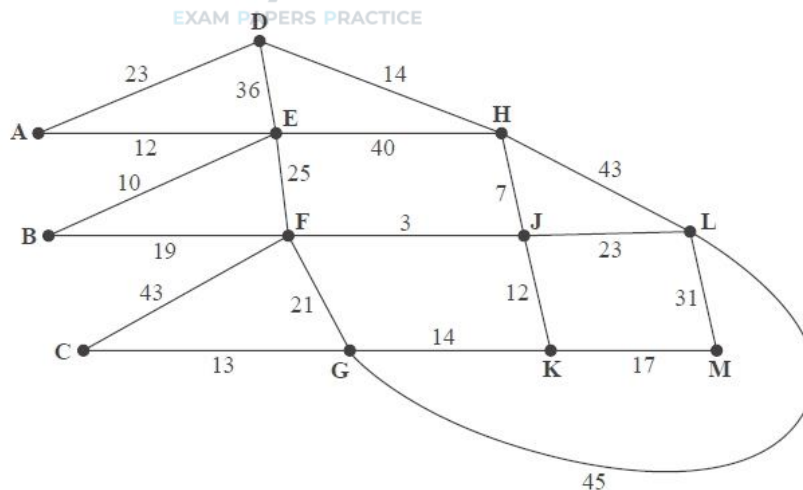


Figure 1 models a network of tracks in a forest that need to be inspected by a park ranger. The number on each arc is the length, in km, of that section of the forest track.

Each track must be traversed at least once and the length of the inspection route must be minimised. The inspection route taken by the ranger must start and end at vertex A.

(a) Use the route inspection algorithm to find the length of a shortest inspection route. State the arcs that should be repeated. You should make your method and working clear.

(5)

(b) State the number of times that vertex J would appear in the inspection route.

(1)

The landowner decides to build two huts, one hut at vertex K and the other hut at a different vertex. In future, the ranger will be able to start his inspection route at one hut and finish at the other. The inspection route must still traverse each track at least once.

(c) Determine where the other hut should be built so that the length of the route is minimised. You must give reasons for your answer and state a possible route and its length.

(4)

Q4.

[The total weight of the network is 384]

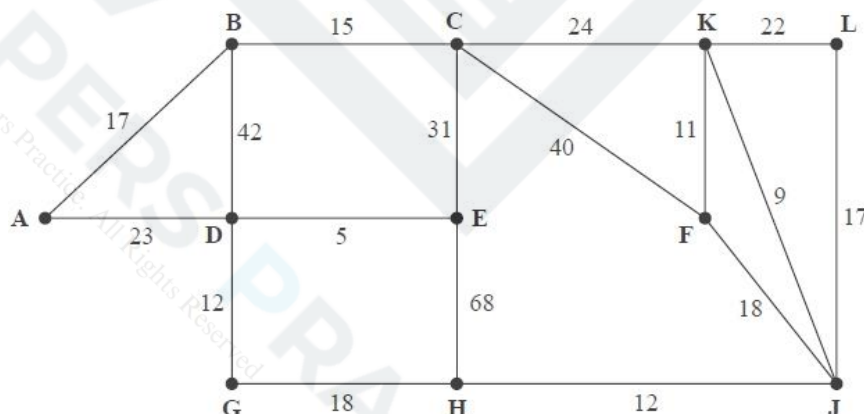


Figure 5

Figure 5 models a network of corridors in an office complex that need to be inspected by a security guard. The number on each arc is the length, in metres, of the corresponding section of corridor.

Each corridor must be traversed at least once and the length of the inspection route must be minimised. The guard must start and finish at vertex A.

(a) Use the route inspection algorithm to find the length of the shortest inspection route. State the arcs that should be repeated. You should make your method and working clear.

(5)

It is now possible for the guard to start at one vertex and finish at a different vertex. An inspection route that traverses each corridor at least once is still required.

(b) Explain why the inspection route should start at a vertex with odd degree.

(2)

The guard decides to start the inspection route at F and the length of the inspection route must still be minimised.

(c) Determine where the guard should finish. You must give reasons for your answer.

(2)

(d) State a possible route and its length.

(2)

END OF HOMEWORK

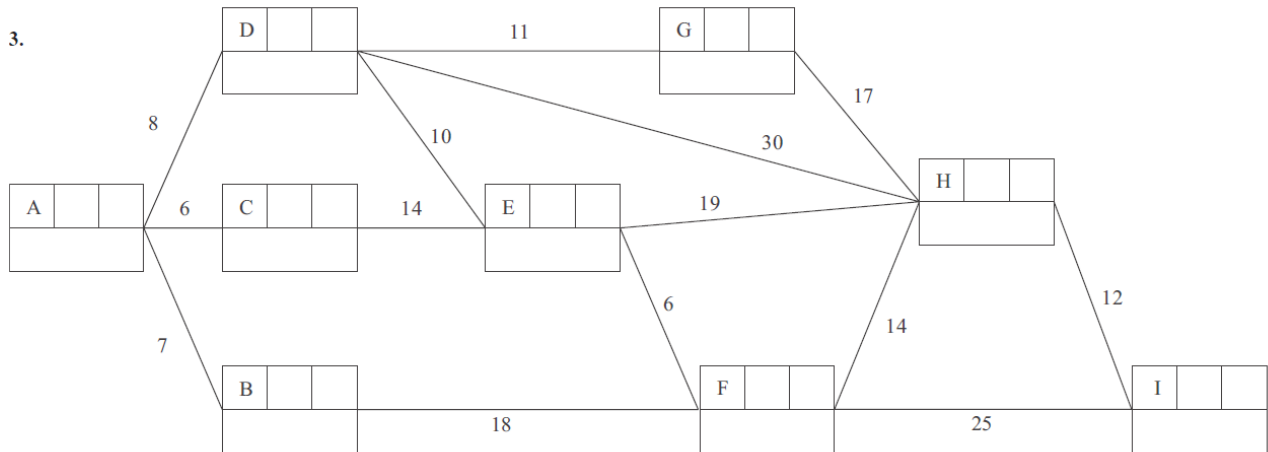
Total = 40 marks

Hand your work in with this sheet. Failure to meet your target grade will result in intervention support.

File this question paper in your file, ready for revision and file inspection.

Answer booklet Q1

3.

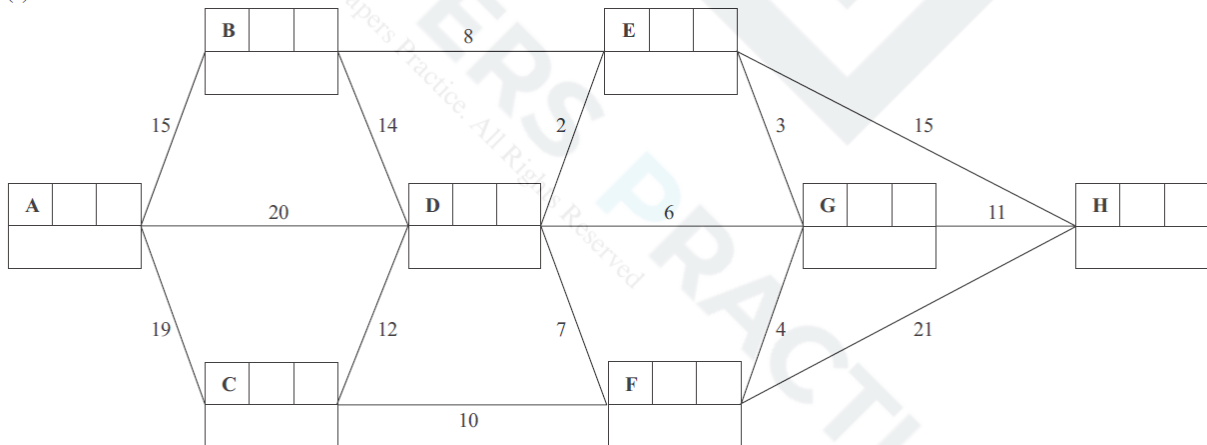


shortest route: _____

length: _____

Answer booklet Q2

3. (a)



Key:

Vertex	Order of labelling	Final value
Working values		

Quickest route: _____

Time taken: _____